

Truthful Online Combinatorial Auction-Based Mechanisms for Task Offloading in Mobile Edge Computing

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Abstract—Mobile edge computation (MEC) is envisioned as a prospective approach for processing the computation-intensive and delay-sensitive tasks of smart mobile devices (SMDs) through offloading them to base stations (BSs) nearby. In fact, efficient task offloading mechanisms are crucial to accomplish an MEC system. The key challenge is to make on-spot decisions upon the arrival of each task and at the same time achieve truthfulness of each SMD. The challenge further escalates, when the unique characteristics of an MEC system, such as locality constraint, delay constraint, etc., are explicitly considered. To solve the challenge, we present a truthful online combinatorial auction-based mechanism (TOCA) for task offloading in an MEC system. Specifically, we first devise the candidate offloading scheme determination algorithm, aiming to determine the candidate offloading schemes of an SMD upon the arrival of its task. Next, we devise the winning offloading scheme selection and pricing algorithm based on the online primal-dual optimization framework, to decide the winning scheme among the SMD's candidate offloading schemes and calculate its payment. By solid theoretical analysis, we verify that TOCA achieves truthfulness, individual rationality and computational efficiency and a smaller competitive ratio. Trace-driven simulation studies validate the effectiveness and efficacy of TOCA.

Index Terms—Mobile edge computing, task offloading, pricing, online auction, truthfulness.

I. INTRODUCTION

IN THE last decade, the number of smart mobile devices (SMDs), e.g., smart phones, wearable devices, smart watches tablets etc., has grown explosively [1], [2], [3]. These SMDs

are typically restricted in their computation capacities and battery because of the physical size constraint, and accordingly usually cannot process computation-intensive tasks correctly [4], [5], [6]. Cloud computing was once regarded as a solution to the restriction of computation capacity in SMDs [7], [8], [9]. Computation-intensive tasks can be offloaded from SMDs to clouds for remote execution. Nevertheless, the cloud computing paradigm is often challenged by a large number of delay-sensitive applications, e.g., interactive games [10] and augmented reality [11]. To alleviate the computational burden of SMDs and to support computation-intensive and delay-sensitive tasks, mobile edge computing (MEC) is introduced [12], [13], [14], [15]. MEC enables SMDs to offload their tasks to adjacent base stations (BSs) integrated with edge servers (ESs) for remote execution so as to shorten the task response delay. In this way, the quality of service (QoS) and energy consumption on SMDs can be significantly improved [16], [17], [18], [19].

In an MEC system, many SMDs have various tasks for remote offloading execution and a network operator owns a number of geo-distributed BSs, each of which owns different amounts of computation and communication resources. In fact, the network operator is unwilling to offer free resources. The resource offering behavior of the network operator needs economic stimuli, e.g., SMDs are required to pay the network operator the cost of the consumed resources. Task offloading in practice often requires a bundle of different computation and communication resources, meanwhile auction has proven to be an efficient resource allocation mechanism [20], [21]. Accordingly, a combinatorial auction is a natural fit here to enable SMDs to efficiently bid for requesting bundles of resources.

However, designing an efficient combinatorial auction-based mechanism for task offloading in an MEC system confronts two critical challenges. *First*, this is essentially an online problem. In fact, SMDs' tasks randomly arrive and leave the MEC system. As a result, it is impossible for the network operator to obtain the information of future task requests before making offloading decisions, which makes the traditional offline auction mechanisms infeasible and requires the design of online auction mechanisms. Furthermore, online auction mechanisms require to make on-spot decisions upon the arrival of each task request, yet still achieving social welfare maximization across the overall system lifespan, which is an NP-hard problem. Thus, it is imperative to devise an approximate online auction mechanism for task offloading in an MEC system.

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Second, an online auction-based mechanism needs to guarantee the desired economical property of truthfulness. In an MEC system, the valuation of each SMD's task is private and unknown to the network operator. As a result, the selfish SMDs have a motivation to improve their utilities through deliberately announcing untruthful valuations. This may deteriorate the long-term gains of the MEC system. Therefore, the key is that the designed online auction-based mechanism can encourage SMDs to tell their truthful valuations. Once SMDs tell untruthful valuations, they will get lower utilities than they tell truthful valuations.

In practice, when striving to design truthful online auction-based mechanisms for task offloading, we have to consider the following characteristics of an MEC system.

- 1) The locality constraint needs to be considered in the task offloading process in an MEC system. Generally, BSs in an MEC system are geographically dispersed, each of which covers a specific geographical region because of the limited communication range. Thus, an SMD can only offload its task to its neighboring BSs and a BS can only serve the SMDs in its coverage.
- 2) The network operator needs to dynamically allocate the communication and computation resources for task offloading. In fact, there are usually multiple adjacent BSs around a task, each of which has different computation and communication resources, meanwhile a task often has some tolerance in delay. Accordingly, there is an urgent need for dynamic resource allocation to determine which BS to acquire resources from, what amounts of resources to obtain from the BS, and which time slots for acquiring these resources from the BS. Since dynamic resource allocation makes full use of network dynamics in subchannel conditions and computation resources, the social welfare can be further improved.

Obviously, the challenges further escalate when the online auction-based mechanism explicitly considers the above characteristics of the MEC system. In reality, once the network operator receiving an SMD's task request, the task offloading in the MEC system can be described as the following essential tasks: *BS selection*, *resource allocation*, *task scheduling* and *payment calculating*. Our target is to pursue the social welfare maximization and guarantee the desired property of truthfulness in these tasks while explicitly considering the above characteristics of the MEC system.

Accordingly, we design a truthful online combinatorial auction-based resource allocation mechanism (TOCA) for task offloading in an MEC system. Different from existing works, TOCA executes in an online manner and achieves the desirable property of truthfulness with explicit consideration of the constraints of locality, deadline and resource. What is more, upon the arrival of an SMD's task, TOCA enables dynamic resource allocation for task offloading, i.e., decisions on which BS to acquire resources from, what amounts of resources to obtain, and which time slots for acquiring the resources, are no longer made randomly a priori, but made dynamically upon receiving the auction result. Besides, the acquired time slots can be consecutive or non-consecutive, thus enabling to offload

different types of tasks. Specifically, upon the arrival of an SMD's task, we first iteratively calculate all its candidate task offloading schemes, which satisfy the constraints of locality, deadline and resource. Next, we set a dynamic unit cost for each type of resources on different BSs, which increases with the consumption of the corresponding type of resources. We then determine the best task offloading scheme, which has the maximum utility. The task is accepted if its best task offloading scheme has positive utility.

The main contributions of this paper are as follows:

- 1) To the best of our knowledge, TOCA is the first truthful online combinatorial auction-based mechanism for task offloading in multiple non-consecutive time slots, which simultaneously considers the constraints of locality, deadline and resource, and enables dynamic resource allocation.
- 2) We propose TOCA which consists of two major components, i.e., the *candidate offloading scheme determination* and the *winning offloading scheme selection and pricing*. For the first component, the network operator helps the current SMD to calculate all its candidate task offloading schemes according to its task request. For the second component, the network operator decides whether the current SMD's task is accepted. If so, the network operator further determines which task offloading scheme of it is accepted and calculates the corresponding payment.
- 3) By theoretical analysis, we prove that TOCA is truthful, individually rational, computationally efficient and achieves a good competitive ratio. Trace-driven simulation studies also validate the effectiveness and efficacy of TOCA.

In the remainder of the paper, we give the system model, online auction framework, computation and communication models, and the problem formulation in Section II. Design and theoretical analysis of TOCA are given in Sections III and IV. Section V presents the performance evaluation of TOCA. Section VI discusses the related works, and Section VII concludes the paper.

II. PRELIMINARY AND PROBLEM FORMULATION

In this section, we first give the system model, online auction framework and the computation and communication models. Next, we formulate the task offloading problem as a mechanism design problem, aiming to maximize the social welfare and achieve the desirable properties. The key notations are summarized in Table I for ease of reference.

A. System Model

We consider an MEC system involving a network operator and a number of SMDs. Let $[Z]$ represent the integer set $\{1, 2, \dots, Z\}$. The MEC system works in a time-slotted manner within T time slots, and the duration of each time slot is ω . The network operator has a collection of J geo-distributed BSs, denoted by $[J]$, each of which is equipped with an ES. The position of each BS $j \in [J]$ is denoted by (X_j, Y_j) , where X_j and Y_j are the longitude and latitude, respectively. Each BS $j \in [J]$ owns K types of computation resources, such as CPU,

TABLE I
KEY NOTATIONS

Notation	Description
T, J, I	number of time slots, BSs and SMDs
(X_j, Y_j)	position of each BS j
A_{jk}	capacity of type k resources in BS j
M	number of VM types
μ_m	computing power of type m VM
H_j	number of wireless subchannels of BS j
θ	bandwidth of a wireless subchannel
ϕ_j	coverage radius of BS j
γ_i	task of SMD i
b_i, v_i	bid price and real valuation of executing task γ_i
(X'_i, Y'_i)	position of SMD i
Π_i	set of the candidate offloading schemes of SMD i
Θ_{in}	set of time slots for executing task γ_i in Π_{in}
j_{in}, m_{in}, l_{in}	specified BS, selected VM type and required number of subchannels to execute task γ_i in Π_{in}
N_i	number of candidate offloading schemes for executing task γ_i
$\hat{t}_{ij}^{cm}, \hat{t}_{ij}^{rx}$	computation time and transmission time of executing task γ_i on BS j
Δ_{ij}	number of time slots of task γ_i executing on BS j
u_i, u_0	utility of SMD i and utility of the network operator
x_{in}	a binary variable, with 1 indicating the accepting of SMD i 's n -th scheme and 0 otherwise
h_{in}^r	total type r resource demand in SMD i 's n -th scheme in time slot t
$\hat{p}_r$	dual variable, indicating the price of type r resource in time slot t
χ_{in}, b'_{in}	total operational cost and net valuation of task offloading scheme Π_{in}
β_r	amount of type r resources consumed in time slot t
$\tilde{A}_r$	available resources of type $r \in [R]$
p_{in}	payment of offloading scheme Π_{in}

RAM, and HDD, and the available resources of each type k on each BS j are A_{jk} . Leveraging the computation resources, network operator can supply M types of virtual machines (VMs) on each BS. Each type m VM consumes η_{mk} units of each type $k \in [K]$ resources, and its corresponding computing power is μ_m CPU cycles per second. Moreover, each BS $j \in [J]$ has H_j orthogonal wireless subchannels for data transferring, each of which owns a bandwidth of θ Mbps, and each SMD is allowed to apply for at most Q subchannels. Due to the limited communication range, each BS j covers a specific geographical region, whose coverage radius is ϕ_j , and can only serve the SMDs in its coverage, denoted as $[J_j]$. In addition, for each BS j , the operational cost of per unit type k computation resource and per subchannel in one time slot is ρ_j^k and ε_j . In order to efficiently use the subchannel resources of a BS, it is assumed that a number of SMDs can access the BS simultaneously, nonetheless, SMDs are required to utilize individual subchannels of the BS.

There are I SMDs, who may arrive in any time slot, and are willing to request the computation and communication resources from BSs to offload their tasks for remote execution. For each SMD $i \in [I]$, its task is denoted as $\gamma_i = (\tau_i, \vartheta_i, \varphi_i, \pi_i, d_i)$,

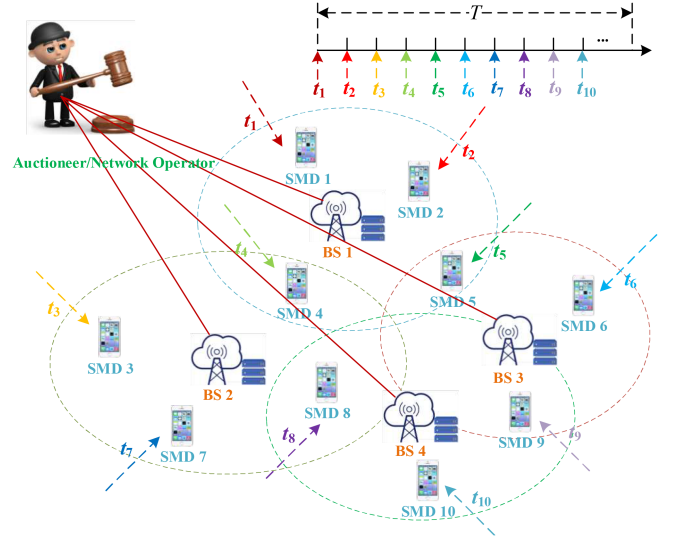


Fig. 1. System model.

where τ_i is the arrival time, ϑ_i is the input data size, φ_i is the required CPU cycles to finish the task, π_i is the output data size, and d_i is the deadline for the task completion. For example, a task of $(1, 20, 50, 1, 5)$ denotes the task arrives at the beginning of time slot index by 1, requires to be completed before the beginning of time slot index by 5, and its input data size, needed CPU cycles and output data size are 20 MB, 50 CPU cycles and 1MB, respectively. In addition, for each SMD i , its position is (X'_i, Y'_i) and it can only offload task γ_i to its adjacent BSs, denoted as $[J_i]$.

B. Online Auction Framework

As shown in Fig. 1, we first give an online auction framework, in which the network operator acting as the auctioneer offers its resources to SMDs, and the SMDs acting as the bidders ask for the computation and communication resources to offload their tasks. The interactions between the network operator and the SMDs are as follows.

- 1) The auctioneer runs the auction in a time-slotted manner within T time slots.
- 2) At the beginning of a certain time slot τ_i , SMD i arrives and submits a bid $B_i = \{\gamma_i, b_i\}$ and its current position (X'_i, Y'_i) to the auctioneer, where b_i is the bid price of offloading task γ_i .
- 3) Upon receiving bid B_i , the auctioneer invokes the *candidate offloading scheme determination* (see Section III-A for details) to calculate its candidate offloading schemes $\Pi_i = (\Theta_{in}, j_{in}, m_{in}, l_{in})_{n \in [N_i]}$ that satisfy the constraints of deadline, locality and resource. Here, Θ_{in} is the set of time slots for executing task γ_i in scheme Π_{in} ; j_{in} , m_{in} and l_{in} are the specified BS, the selected VM type and the required number of subchannels to offload task γ_i in scheme Π_{in} ; N_i is the number of candidate offloading schemes for task γ_i .
- 4) The auctioneer invokes the *winning offloading scheme selection and pricing* (see Section III-B for details) to

decide whether to accept task γ_i . If so, the auctioneer needs to further decide which offloading scheme wins and calculate the corresponding payment. Then, the auctioneer announces the auction result to SMD i .

- 5) SMD i offloads its task according to the related winning offloading scheme.
- 6) The auctioneer charges an amount of money p_i from SMD i for its task offloading.

Note that, in our online model, there exist multiple candidate offloading schemes for a task, but at most one offloading scheme can be accepted. In this way, the network operator can dynamically allocate computation and communication resources to SMDs according to their winning offloading schemes. Moreover, since SMDs are selfish and strategic, the bid price b_i may not equal the real valuation v_i .

C. Computation and Communication Models

For any task γ_i , when SMD i selects type m_{ij} VM from BS j to offload its task, its computation time is defined as

$$\hat{t}_{ij}^{cm} = \frac{\varphi_i}{\mu_{m_{ij}}} \quad (1)$$

where m_{ij} denotes the selected VM type of BS j on which task γ_i is offloaded.

Next, the communication model is introduced for input data uploading and output data downloading. Let l_{ij} be the number of subchannels acquired by SMD i from BS j . Thus, by Shannon formula, the achievable data transmission rate between SMD i and BS j can be formulated as

$$q_{ij} = l_{ij} \theta \log_2 \left(1 + \frac{\delta_i \sigma_{ij}}{\varpi_i} \right) \quad (2)$$

where δ_i represents the transmission power of SMD i , σ_{ij} denotes the channel gain between SMD i and BS j , and ϖ_i represents the background noise. Further, the total transmission time of input and output data between SMD i and BS j is

$$\hat{t}_{ij}^{tx} = \frac{\vartheta_i + \pi_i}{q_{ij}} \quad (3)$$

Thus, the number of time slots, i.e., Δ_{ij} , that SMD i requires to finish task γ_i with type m_{ij} VM and l_{ij} number of subchannels, can be calculated by

$$\Delta_{ij} = \left\lceil \frac{\hat{t}_{ij}^e + \hat{t}_{ij}^{tx}}{\omega} \right\rceil \quad (4)$$

where $\lceil x \rceil$ denotes the ceil function which returns the smallest integer no less than x .

D. Problem Formulation

Let x_{ij} and y_{ijt} be binary variables. $x_{ij} = 1$ if task γ_i is offloaded to BS j , and 0 otherwise; $y_{ijt} = 1$ if task γ_i is offloaded to BS j at the beginning of time slot t , and 0 otherwise.

Definition 1 (SMD Utility): The utility of SMD i equals the real valuation less the payment, defined as:

$$u_i = \sum_{j \in [J_i]} x_{ij} v_i - p_i \quad (5)$$

Definition 2 (Network Operator Utility): The network operator's utility is the aggregate payments of all the SMDs minus the overall operational costs of all the BSs, defined as:

$$u_0 = \sum_{i \in [I]} p_i - \sum_{i \in [I]} \sum_{j \in [J_i]} x_{ij} \Delta_{ij} \left(\sum_{k \in [K]} \eta_{m_{ij}k} \rho_j^k + l_{ij} \varepsilon_j \right) \quad (6)$$

Definition 3 (Social Welfare): Social welfare equals the utilities of all the participants, including the network operator and all the SMDs, i.e.,

$$SW = \sum_{i \in [I]} \sum_{j \in [J_i]} x_{ij} \left(v_i - \Delta_{ij} \left(\sum_{k \in [K]} \eta_{m_{ij}k} \rho_j^k + l_{ij} \varepsilon_j \right) \right) \quad (7)$$

Definition 4 (Competitiveness in Social Welfare): An online auction is ξ -competitive in social welfare if the ratio of the social welfare obtained by offline optimal auction to the social welfare acquired by an online auction is upper-bounded by ξ in any cases.

Definition 5 (Truthfulness): An auction mechanism achieves truthfulness if for any SMD, reporting the true valuation forms its dominant strategy, regardless of other SMDs' bids.

Definition 6 (Individual Rationality): An auction mechanism achieves individual rationality if each participant always acquires a non-negative utility.

Definition 7 (Computational Efficiency): An auction mechanism achieves computational efficiency if the task offloading algorithm and pricing algorithm can be done in polynomial time.

According to the above definitions, the social welfare maximization (SWM) problem can be formulated as an integer linear program (ILP) as follows.

$$\max SW = \sum_{i \in [I]} \sum_{j \in [J_i]} x_{ij} \left(b_i - \Delta_{ij} \left(\sum_{k \in [K]} \eta_{m_{ij}k} \rho_j^k + l_{ij} \varepsilon_j \right) \right) \quad (8)$$

subject to:

$$y_{ijt} t \leq x_{ij} d_i, \forall i \in [I], j \in [J_i], t \in [T] : t \geq \tau_i \quad (8.1)$$

$$x_{ij} \Delta_{ij} \leq \sum_{t \in [T] : t \geq \tau_i} y_{ijt}, \forall i \in [I], j \in [J_i] \quad (8.2)$$

$$\sum_{i \in [I_j] : t \geq \tau_i} y_{ijt} \eta_{m_{ij}k} \leq A_{jk}, \forall j \in [J], k \in [K], t \in [T] \quad (8.3)$$

$$\sum_{i \in [I_j] : t \geq \tau_i} y_{ijt} l_{ij} \leq H_j, \forall j \in [J], t \in [T] \quad (8.4)$$

$$\sum_{j \in [J_i]} x_{ij} \leq 1, \forall i \in [I] \quad (8.5)$$

$$x_{ij} \in \{0, 1\}, \forall i \in [I], j \in [J_i] \quad (8.6)$$

$$y_{ijt} \in \{0, 1\}, \forall i \in [I], j \in [J_i], t \in [T] : t \geq \tau_i \quad (8.7)$$

$$m_{ij} \in [A_j], \forall i \in [I], j \in [J_i] \quad (8.8)$$

$$l_{ij} \in [H_j], \forall i \in [I], j \in [J_i] \quad (8.9)$$

The objective of the SWM problem is to maximize the social welfare, which can prompt the efficiency of the whole eco-system and make everyone have a higher probability to be happy in the long run. Constraint (8.1) guarantees that a task is offloaded after its arrival time and finished before its deadline. Constraint (8.2) ensures that the number of allocated time slots is sufficient for offloading the task. Constraint (8.3) expresses the capacity limit of each type of computation resource on each BS in each time slot. Constraint (8.4) indicates the number limit of subchannels on each BS in each time slot. Constraint (8.5) indicates if SMD i wins, it can offload its task to one BS only. Constraints (8.6), (8.7), (8.8) and (8.9) ensure the restriction on the decision variables. Noting that, to enable flexible task offloading, the allocated time slots for offloading a task don't need to be consecutive. That is, an SMD i 's task γ_i can be offloaded on BS j at any time slot as long as the total offloading time meets Δ_{ij} before the deadline. In addition, Δ_{ij} is not a decision variable which can be calculated according to (1), (2), (3) and (4).

Even given complete future information, i.e., in the offline case, ILP (8) ignoring constraints (8.1) and (8.2) is yet an NP-hard problem, which is equivalent to the well-known knapsack problem. The challenge will further escalate while considering the constraints of deadline and locality and pursuing online decision making. To solve the challenges, we first present a novel framework to deal with the constraints of deadline and locality, and then employ primal-dual technique to reformulate the original ILP (8) into an equivalent compact-exponential ILP with packing structure.

$$\begin{aligned} \max SW = & \sum_{i \in [I]} \sum_{n \in [N_i]} x_{in} \\ & \left(b_i - \sum_{t \in \Theta_{in}} \left(\sum_{k \in [K]} \eta_{m_{in}k} \rho_{j_{in}}^k + l_{in} \varepsilon_{j_{in}} \right) \right) \end{aligned} \quad (9)$$

subject to:

$$\sum_{\substack{i \in [I_j], n \in [N_i]: \\ j=j_{in}, t \in \Theta_{in}}} x_{in} \eta_{m_{in}k} \leq A_{jk}, \forall j \in [J], k \in [K], t \in [T] \quad (9.1)$$

$$\sum_{\substack{i \in [I_j], n \in [N_i]: \\ j=j_{in}, t \in \Theta_{in}}} x_{in} l_{in} \leq H_j, \forall j \in [J], t \in [T] \quad (9.2)$$

$$\sum_{n \in [N_i]} x_{in} \leq 1, \forall i \in [I] \quad (9.3)$$

$$x_{in} \in \{0, 1\}, \forall i \in [I], n \in [N_i] \quad (9.4)$$

In the above compact-exponential ILP, Π_i is the set of N_i candidate offloading schemes for task γ_i . A candidate scheme is a vector $\Pi_{in} = (\Theta_{in}, j_{in}, m_{in}, l_{in})$, which can be easily transformed to the equivalent vector $(x_{ij}, y_{ijt}, m_{ij}, l_{ij})$ satisfying

constraints (8.1) and (8.2). A binary variable x_{in} equals 1 if SMD i 's n -th scheme is accepted, and 0 otherwise.

We view each computation and communication resource on all BSs as different resource types. Thus, there are a total of $R = JK + J$ types of resources, i.e., JK types of computation resources and J types of subchannel resources on all the BSs. Let $\tilde{A}_r$ be the available resources of type $r \in [R]$, and h_{in}^{rt} be the aggregated resource demand of type r in SMD i 's n -th scheme in time slot t . For instance, for offloading scheme Π_{in} , if resource r corresponds to computation resource k , $h_{in}^{rt} = \eta_{m_{in}k}$; if resource r corresponds to subchannel resource, $h_{in}^{rt} = l_{in}$. In addition, the overall operational cost of task offloading scheme Π_{in} can be readily computed, i.e., $\chi_{in} = \sum_{r \in [R]} \sum_{t \in \Theta_{in}} h_{in}^{rt} \hat{\chi}_r$, where $\hat{\chi}_r$ is the operational cost of per unit resource of type r . We define the net valuation of offloading scheme Π_{in} as its valuation minus the corresponding operational cost, i.e., $b'_{in} = b_i - \chi_{in}$. Thus, the social welfare is equal to the aggregated net valuations of all the winning SMDs. Based on the above analysis, we reformulate the ILP (9) as follows.

$$\max SW = \sum_{i \in [I]} \sum_{n \in [N_i]} b'_{in} x_{in} \quad (10)$$

subject to:

$$\sum_{\substack{i \in [I_j], n \in [N_i]: \\ t \in \Theta_{in}}} x_{in} h_{in}^{rt} \leq \tilde{A}_r, \forall r \in [R], t \in [T] \quad (10.1)$$

$$\sum_{n \in [N_i]} x_{in} \leq 1, \forall i \in [I] \quad (10.2)$$

$$x_{in} \in \{0, 1\}, \forall i \in [I], n \in [N_i] \quad (10.3)$$

It is obvious that constraint (10.1) is equivalent to constraints (9.1) and (9.2), and constraint (10.2) is equivalent to constraint (9.3).

We relax the integrality constraints of $x_{in} \in \{0, 1\}$ to $x_{in} \geq 0$, and introduce dual variables $\hat{p}_{rt}$ and u_i to constraints (10.1) and (10.2). Thus, the dual problem of the relaxation of ILP (10) is formulated as:

$$\min \sum_{t \in [T]} \sum_{r \in [R]} \tilde{A}_r \hat{p}_{rt} + \sum_{i \in [I]} u_i \quad (11)$$

subject to:

$$u_i \geq b'_{in} - \sum_{r \in [R]} \sum_{t \in \Theta_{in}} h_{in}^{rt} \hat{p}_{rt}, \forall i \in [I], n \in [N_i] \quad (11.1)$$

$$\hat{p}_{rt} \geq 0, u_i \geq 0, \forall i \in [I], r \in [R], t \in [T] \quad (11.2)$$

ILP (8), ILP (9) and ILP (10) have the same optimal objective value. To resolve ILP (10), complete knowledge is needed throughout the system lifetime. Nevertheless, our mechanism needs to operate in an online manner, making on-spot decisions without depending on the future information. Therefore, we employ the primal-dual technique which calculates the primal solution according to the dual variables.

III. DESIGN OF TOCA

In this section, we first design the *candidate offloading scheme determination* to calculate the set of candidate offloading scheme for an SMD's task. Then, we design the *winning offloading scheme selection and pricing* to determine whether to accept the SMD's task or not. If the task is accepted, we select the winning offloading scheme that maximizes the SMD's utility. Further, we calculate the payment of the winning offloading scheme based on the demand-supply relationship in the MEC system. Using the above algorithms as building blocks, we present the design of our proposed TOCA.

A. Candidate Offloading Scheme Determination

In this section, we present the *candidate offloading scheme determination*, shown as Algorithm 1. Upon the arrival of an SMD's task, we use the *candidate offloading scheme determination* to calculate the set of its candidate offloading schemes, each of which satisfies the constraints of locality, deadline and resource. The *candidate offloading scheme determination* has eleven input parameters: $\gamma_i, \{(X'_i, Y'_i)\}, \{(X_j, Y_j)\}, [J], [M], \{\mu_m\}, \{H_j\}, \{\phi_j\}, \theta, \{\hat{p}_{rt}\}$ and $\{\beta_{rt}\}$, and one output parameter: Π_i (see Table I for details). First, we set $[J_i] = \emptyset$ and $n = 0$ (Line 1), and then compute the set of adjacent BSs on which SMD i can offload its task, i.e., $[J_i]$ (Lines 2-4). Next, we iteratively compute the set of candidate offloading schemes, i.e., Π_i , each of which satisfies the deadline constraint, locality constraint and resource constraint (Lines 5-17). Specifically, we determine a number of time slots within $[\tau_i, d_i]$ for task γ_i as follows: we first calculate the task computation time $\hat{t}_{ij}^{cm}$ (Line 7) and data transmission time $\hat{t}_{ij}^{tx}$ (Lines 8-9), respectively. Next, we calculate the required number of time slots, i.e., $\Delta_{ij} = \lceil (\hat{t}_{ij}^{cm} + \hat{t}_{ij}^{tx}) / \omega \rceil$ (Line 10). Then, we choose Δ_{ij} time slots with minimum $(c(t)), \beta_{jkt} + \eta_{mk} \leq A_{jk}, \forall k \in [K]$ and $\beta_{jt}^{sub} + l \leq H_j$, within $[\tau_i, d_i]$, save these time slots into the set Θ_{in} (Lines 11-14), where β_{jkt} and β_{jt}^{sub} denote the amount of type k resources and the number of subchannels consumed in time slot t , and $\hat{p}_{jkt}$ and $\hat{p}_{jt}^{sub}$ denote the unit price of type k resource and the unit price of subchannel in time slot t .

B. Winning Offloading Scheme Selection and Pricing

The basic idea for the design of the *winning offloading scheme selection and pricing* is as follows. We employ the primal-dual approach to the ILP (10) and its dual (11). Upon arrival of task γ_i , a collection of primal variables $x_{in}, \forall n \in [N_i]$ and the related dual constraints $(b'_{in} - \sum_{r \in [R]} \sum_{t \in \Theta_{in}} h_{in}^{rt} \hat{p}_{rt}, \forall n \in [N_i])$ emerge, where $N_i = |\Pi_i|$. According to the complementary slackness in the Karush-Kuhn-Tucker [22], primal variable x_{in} holds 0 unless the related dual constraint (11.1) is tight. Since the dual variable $u_i \geq 0$, u_i can be set to the maximum of zero and the right-hand side (RHS) of (11.1), i.e.,

$$u_i = \max_{n \in [N_i]} \left\{ 0, \max \left\{ b'_{in} - \sum_{r \in [R]} \sum_{t \in \Theta_{in}} h_{in}^{rt} \hat{p}_{rt} \right\} \right\}. \quad (12)$$

Algorithm 1: Candidate Offloading Scheme Determination.

Input: $\gamma_i, \{(X'_i, Y'_i)\}, \{(X_j, Y_j)\}, [J], \{A_j\}, \{H_j\}, \{\phi_j\}, \theta, \{\mu_m\}, \{\hat{p}_{rt}\}, \{\beta_{rt}\}$.
Output: Π_i .
1: $[J_i] = \emptyset, n = 0$;
2: **for** each $j \in [J]$ **do**
3: **if** $\sqrt{(X'_i - X_j)^2 + (Y'_i - Y_j)^2} \leq \phi_j$ **then**
4: $[J_i] = [J_i] \cup \{j\}$;
5: **for** each $j \in [J_i]$ **do**
6: **for** each $m \in [M], l \in [Q]$ **do**
7: $\hat{t}_{ij}^{cm} = \varphi_i / \mu_m$;
8: $q_{ij} = l \theta \log_2(1 + (\delta_i \sigma_{ij} / \varpi_i))$;
9: $\hat{t}_{ij}^{tx} = (\vartheta_i + \pi_i) / q_{ij}$;
10: $\Delta_{ij} = \lceil (\hat{t}_{ij}^{cm} + \hat{t}_{ij}^{tx}) / \omega \rceil$;
11: set $\{\hat{p}_{jkt}\}$ and $\{\hat{p}_{jt}^{sub}\}$ according to $\{\hat{p}_{rt}\}$;
12: set $\{\beta_{jkt}\}$ and $\{\beta_{jt}^{sub}\}$ according to $\{\beta_{rt}\}$;
13: $c(t) = \sum_{k \in [K]} \eta_{mk} \hat{p}_{jkt} + l \hat{p}_{jt}^{sub}$;
14: choose Δ_{ij} time slots with minimum $(c(t))$,
 $\beta_{jkt} + \eta_{mk} \leq A_{jk}, \forall k \in [K]$ and
 $\beta_{jt}^{sub} + l \leq H_j$, within $[\tau_i, d_i]$, save these time
 slots into the set Θ_{in} ;
15: $j_{in} = j, m_{in} = m, l_{in} = l$;
16: $\Pi_{in} = (\Theta_{in}, j_{in}, m_{in}, l_{in})$;
17: $n = n + 1$;
18: **return** Π_i ;

Therefore, the network operator accepts task γ_i if $u_i > 0$, and offloads it based on the scheme which maximizes RHS of constraint (11.1); otherwise, task γ_i is rejected.

If the dual variable $\hat{p}_{rt}$ is interpreted as the unit price of type r resources in time slot t , then $\sum_{r \in [R]} \sum_{t \in \Theta_{in}} h_{in}^{rt} \hat{p}_{rt}$ indicates the aggregated costs of offloading task γ_i with its n -th scheme. The RHS of (11.1) is the utility of SMD i with its n -th scheme. As a result, the assignment of u_i in (12) effectively maximizes the utility of SMD i , which is a critical step to achieve truthfulness and social welfare maximization. However, the challenge in online mechanism design lies in how to update the dual variables. Specifically, we need to devise a unit price updating function that guarantees the network operator does not accept too many bids with low valuations at the beginning, resulting in rejecting the bids with high valuations arriving in the future. In this paper, the exponential function is used as the basic expression to describe the market strategy. Let β_{rt} be the amount of type r resources consumed in time slot t . Thus, the unit price updating function $\hat{p}_{rt}$ can be defined as:

$$\hat{p}_{rt} = \frac{L_r}{e} \left(\frac{e U_r}{L_r} \right)^{\frac{\beta_{rt}}{A_r}}, \forall r \in [R], \forall t \in [T] \quad (13)$$

where $L_r = \min_{i \in [I], n \in [N_i]} \frac{b'_{in}}{\sum_{t \in \Theta_{in}} h_{in}^{rt}}$ and $U_r = \max_{i \in [I], n \in [N_i]} \frac{b'_{in}}{\sum_{t \in \Theta_{in}} h_{in}^{rt}}$.

In our unit price updating function, $\hat{p}_{rt}$ begins at L_r/e , exponentially grows before β_{rt} approaches A_r , and equals U_r when

Algorithm 2: Winning Offloading Scheme Selection and Pricing.

Input: $\Pi_i, \{b'_{in}\}, \{\tilde{A}_r\}, \{\hat{p}_{rt}\}, \{\beta_{rt}\}$.

Output: $\{x_{in}\}, \{p_{in}\}$.

```

1:  $N_i = |\Pi_i|$ ;
2:  $x_{in} = 0, c_{in} = 0, p_{in} = 0, u_i = 0, \forall n \in [N_i]$ ;
3: for each  $n \in [N_i]$  do
4:   for each  $t \in \Theta_{in}$  do
5:      $c_{in} = c_{in} + \sum_{r \in [R]} h_{in}^{rt} \hat{p}_{rt}$ ;
6:      $u_{in} = b'_{in} - c_{in}$ ;
7:      $n^* = \arg \max_{n \in [N_i]} \{u_{in}\}$ ;
8:   if  $u_{in^*} > 0$  then
9:      $x_{in^*} = 1, p_{in^*} = c_{in^*}, u_i = u_{in^*}$ ;
10:  return  $\{x_{in}\}, \{p_{in}\}$ ;

```

$\beta_{rt} = \tilde{A}_r$. In the case of $\hat{p}_{rt} = U_r$, the network operator will not allocate any type r resources to any SMD.

According to the above analysis, we present the *winning offloading scheme selection and pricing*, described as Algorithm 2. It has five input parameters: $\Pi_i, \{b'_{in}\}, \{\tilde{A}_r\}, \{\hat{p}_{rt}\}, \{\beta_{rt}\}$, and two output parameters: $\{x_{in}\}$ and $\{p_{in}\}$ (see Table I for details). In the *winning offloading scheme selection and pricing*, we first initialize N_i to $|\Pi_i|$, i.e., the size of the set Π_i (Line 1), and then we respectively set x_{in}, c_{in} and p_{in} to 0, 0 and 0 for each $n \in [N_i]$ (Line 2). Next, for scheme Π_{in} , we calculate its c_{in} and further compute its utility $u_{in} = b'_{in} - c_{in}$ (Lines 3-6). Then, we choose scheme Π_{in^*} which maximizes SMD i 's utility (Line 7). If SMD i acquires the positive utility, $x_{in^*} = 1$ and the corresponding payment is $p_{in^*} = c_{in^*}$ (Lines 8-9).

C. TOCA

TOCA in Algorithm 3 is our proposed online algorithm, which can call the algorithms designed in the previous sections as its sub-routines. Specifically, we first define the unit price function and initialize the variables of $\hat{p}_{rt}$ and β_{rt} (Lines 1-2). Upon the arrival of task γ_i , we invoke the *candidate offloading scheme determination* to calculate the set of its candidate offloading schemes Π_i (Line 4). We calculate h_{in}^{rt} for each scheme Π_{in} (Lines 6-11). That is, if resource r corresponds to computation resource k , $h_{in}^{rt} = \eta_{m_{in}k}$ (Line 9); if resource r corresponds to subchannel resource, $h_{in}^{rt} = l_{in}$ (Line 11). Then, we calculate the operational cost and net valuation of each scheme Π_{in} (Lines 12-14). Further, we invoke the *winning offloading scheme selection and pricing* to determine the winning scheme for the task and calculate its corresponding payment (Line 15). If there exists $x_{in'}$ that equals 1, we accept task γ_i , offload it according to scheme $\Pi_{in'}$ and charge $p_{in'}$ for offloading task γ_i (Lines 16-18). At the same time, we update the usage of resources β_{rt} and the price of $\hat{p}_{rt}$ (Lines 19-20). Otherwise, we reject task γ_i (Line 22).

IV. THEORETICAL ANALYSIS

A. Truthfulness and Individual Rationality

Theorem 1: TOCA achieves truthfulness.

Algorithm 3: TOCA.

Input: $\{(X_j, Y_j)\}, \{\tilde{A}_r\}, \{\eta_{mk}\}, \{\mu_m\}, \{\phi_j\}, \theta$.

```

1: Define function  $\hat{p}_{rt}$  according to (13);
2: Initialize  $\hat{p}_{rt} = 0, \beta_{rt} = 0, \forall t \in [T], \forall r \in [R]$ ;
3: while the arrival of SMD  $i$ 's task  $\gamma_i$  do
4:   invoke the candidate offloading scheme determination to obtain  $\Pi_i$ ;
5:    $N_i = |\Pi_i|$ ;
6:   for each  $n \in [N_i]$  do
7:     for each  $r \in [R]$  do
8:       if  $r \in [K]$  then
9:          $h_{in}^{rt} = \eta_{m_{in}k}$ ;
10:      else
11:         $h_{in}^{rt} = l_{in}$ ;
12:      for each  $t \in \Theta_{in}$  do
13:         $\chi_{in} = \sum_{r \in [R]} \sum_{t \in \Theta_{in}} h_{in}^{rt} \hat{p}_{rt}$ ;
14:         $b'_{in} = b_i - \chi_{in}$ ;
15:   invoke the winning offloading scheme selection and pricing to obtain  $\{x_{in}\}$  and  $\{p_{in}\}$ ;
16:   if  $\exists n' \in [N_i], x_{in'} = 1$  then
17:     accept task  $\gamma_i$  and offload it according to scheme  $\Pi_{in'}$ ;
18:     charge  $p_{in'}$  for offloading task  $\gamma_i$ ;
19:      $\beta_{rt} = \beta_{rt} + h_{in'}^{rt}, \forall r \in [R], t \in \Theta_{in}$ ;
20:      $\hat{p}_{rt} = \frac{L_r}{e} \left( \frac{eU_r}{L_r} \right)^{\frac{\beta_{rt}}{A_r}}, \forall r \in [R], t \in \Theta_{in}$ ;
21:   else
22:     reject task  $\gamma_i$ ;

```

Proof: Upon the arrival of task γ_i , the payment that SMD i needs to pay to the network operator if its task is accepted, depends only on the amount of the used resources and SMD i 's request. It is independent of SMD i 's bid price. Therefore, SMD i cannot increase the utility through untruthful bidding as the utility is equal to the valuation less the payment. Moreover, TOCA always chooses the scheme with the maximum utility among all the candidate schemes for task γ_i . Thus, truthful bidding ensures that each SMD acquires the maximum utility in TOCA. $\square$

Theorem 2: TOCA achieves individual rationality.

Proof: Since $u_i = \max\{0, \max\{b'_{in} - \sum_{r \in [R]} \sum_{t \in \Theta_{in}} h_{in}^{rt} \hat{p}_{rt}\}\}$. If $u_i = 0$, the network operator rejects task γ_i , and the SMD's utility is $u_i = 0$. If $u_i > 0$, task γ_i wins. Therefore, the SMD's utility is always no less than 0, which guarantees TOCA is individually rational. $\square$

B. Correctness and Computational Efficiency

Lemma 1: The *candidate offloading scheme determination* is computationally efficient.

Proof: The *candidate offloading scheme determination* consists of two *for* loops. The first *for* loop (Lines 2-4) requires the polynomial time $O(J)$. The second *for* loop (Lines 5-17) requires the polynomial time $O(JMQT \log T)$. Accordingly, the *candidate offloading scheme determination* achieves computational efficiency. $\square$

Lemma 2: The winning offloading scheme selection and pricing is computationally efficient.

Proof: The winning offloading scheme selection and pricing mainly consists of a for loop and a maximizing operation. The for loop (Lines 3-6) requires the polynomial time $O(N'T)$, where $N' = \max_{i \in [I]} \{N_i\}$. The maximizing operation (Line 7) requires the polynomial time $O(N')$. Accordingly, the winning offloading scheme selection and pricing achieves computational efficiency. $\square$

Theorem 3: TOCA generates a feasible solution to ILP (9), ILP (10) and DLP (11) in polynomial time.

Proof. Correctness: Constraint (10.1) holds as line 14 in the candidate offloading scheme determination guarantee each candidate offloading scheme Π_{in} cannot violate resource capacity restriction. Constraint (10.2) holds as only one offloading scheme per task wins according to line 7 in the winning offloading scheme selection and pricing. Thus, TOCA generates a feasible solution to ILP (10). Moreover, since ILP (9) is equivalent to ILP (10), x_{in} is also a feasible solution to ILP (9). In addition, since the winning offloading scheme selection and pricing sets $u_i = 0$ if $b'_{in} \leq \sum_{r \in [R]} \sum_{t \in \Theta_{in}} h_{in}^{rt} \hat{p}_{rt}$, and $u_i = b'_{in} - \sum_{r \in [R]} \sum_{t \in \Theta_{in}} h_{in}^{rt} \hat{p}_{rt}$ otherwise, TOCA generates a feasible solution to DLP (11).

Computational efficiency: TOCA consists of two sub-routines, i.e., the candidate offloading scheme determination and the winning offloading scheme selection and pricing, and two for loop. By Lemmas 1 and 2, the candidate offloading scheme determination and the winning offloading scheme selection and pricing require the polynomial time $O(JMQT \log T)$ and $O(N'T)$, respectively. Furthermore, the first for loop (Lines 6-11) and the second for loop (Lines 12-14) require the polynomial time $O(N'R)$ and $O(N'TR)$, respectively. Thus, TOCA achieves computational efficiency. $\square$

C. Competitive Ratio

In the following, we analyze the competitive ratio of TOCA. In order to calculate the competitive ratio, we make the following assumption, which holds true in the real world.

Assumption 1: The consumed resources of type r for offloading one task is much smaller than the resource capacity of type r on a BS, i.e., $h_{in}^{rt} \ll \tilde{A}_r, \forall r \in [R]$.

Let $\hat{p}_{rt}^i$ be the price of type r resource after processing task γ_i , and β_{rt}^i be the amounts of type r resources consumed after processing task γ_i . $\hat{p}_{rt}^i$ and β_{rt}^i will not change when task γ_i is not accepted, thus the following derivation is under the condition that task γ_i is accepted in its n -th task offloading scheme Π_{in} .

Lemma 3: If $\xi = \max_{r \in [R]} \xi_r$, then we have $\hat{p}_{rt}^{i-1}(\beta_{rt}^i - \beta_{rt}^{i-1}) \geq \frac{1}{\xi} \tilde{A}_r(\hat{p}_{rt}^i - \hat{p}_{rt}^{i-1})$, where $\xi_r = \ln(U_r/L_r) + 1, \forall i \in [I], \forall r \in [R], \forall t \in \Theta_{in}$.

Proof: The derivative of the price function in (13) is:

$$d\hat{p}_{rt} = \frac{L_r}{e\tilde{A}_r} \left(\frac{eU_r}{L_r} \right)^{\frac{\beta_{rt}}{\tilde{A}_r}} \left(\ln \left(\frac{U_r}{L_r} \right) + 1 \right) d\beta_{rt}$$

Then, we have

$$\frac{1}{\xi_r} \tilde{A}_r d\hat{p}_{rt} = \frac{1}{\left(\ln \left(\frac{U_r}{L_r} \right) + 1 \right)} \tilde{A}_r \frac{L_r}{e\tilde{A}_r} \left(\frac{eU_r}{L_r} \right)^{\frac{\beta_{rt}}{\tilde{A}_r}} \left(\ln \left(\frac{U_r}{L_r} \right) + 1 \right)$$

$$d\beta_{rt} = \hat{p}_{rt} d\beta_{rt}.$$

Since $\xi = \max_{r \in [R]} \xi_r$, we have

$$\hat{p}_{rt} d\beta_{rt} \geq \frac{1}{\xi} \tilde{A}_r d\hat{p}_{rt} \quad (14)$$

Based on the Assumption 1, i.e., $h_{in}^{rt} \ll \tilde{A}_r, \forall r \in [R], \beta_{rt}^i - \beta_{rt}^{i-1}$ can be expressed as $d\beta_{rt}$. Further, we can get $d\hat{p}_{rt} = \hat{p}'_{rt}(\beta_{rt}) d\beta_{rt} = \hat{p}_{rt}^i - \hat{p}_{rt}^{i-1}$. Thus, (14) can be rewritten as

$$\hat{p}_{rt}^{i-1}(\beta_{rt}^i - \beta_{rt}^{i-1}) \geq \frac{1}{\xi} \tilde{A}_r(\hat{p}_{rt}^i - \hat{p}_{rt}^{i-1}) \quad (15)$$

Let V_1 and V_2 be optimal objective values of ILP (9) and ILP (10), respectively. We can easily have $V_1 = V_2$. Let P_i and D_i denote the objective values of ILP (10) and its DLP (11) obtained by our proposed TOCA after accepting task γ_i . Let P_0 and D_0 be the initial objective values, and $P_0 = D_0 = 0$. Let P_I and D_I represent the objective values of ILP (10) and its dual problem (11) obtained by our proposed TOCA after it has handled all the SMDs' tasks.

Lemma 4: If $\hat{p}_{rt}^{i-1}(\beta_{rt}^i - \beta_{rt}^{i-1}) \geq \frac{1}{\xi} \tilde{A}_r(\hat{p}_{rt}^i - \hat{p}_{rt}^{i-1})$ holds, then TOCA ensures that $P_i - P_{i-1} \geq \frac{1}{\xi}(D_i - D_{i-1})$ for each SMD $i \in [I]$.

Proof: Since TOCA make constraint (11.1) tight when selecting scheme Π_{in} , $b'_{in} = u_i + \sum_{t \in \Theta_{in}} \sum_{r \in [R]} \hat{p}_{rt}^{i-1}(\beta_{rt}^i - \beta_{rt}^{i-1})$. By the definition of P_i and D_i , we have

$$P_i - P_{i-1} = b'_{in} = u_i + \sum_{t \in \Theta_{in}} \sum_{r \in [R]} \hat{p}_{rt}^{i-1}(\beta_{rt}^i - \beta_{rt}^{i-1}) \quad (16)$$

$$D_i - D_{i-1} = u_i + \sum_{t \in \Theta_{in}} \sum_{r \in [R]} \tilde{A}_r(\hat{p}_{rt}^i - \hat{p}_{rt}^{i-1}) \quad (17)$$

Summing up the relationship of $\hat{p}_{rt}^{i-1}(\beta_{rt}^i - \beta_{rt}^{i-1}) \geq \frac{1}{\xi} \tilde{A}_r(\hat{p}_{rt}^i - \hat{p}_{rt}^{i-1})$ for each $\forall r \in [R], \forall t \in \Theta_{in}$, we get

$$\sum_{t \in \Theta_{in}} \sum_{r \in [R]} \hat{p}_{rt}^{i-1}(\beta_{rt}^i - \beta_{rt}^{i-1}) \geq \frac{1}{\xi} \sum_{t \in \Theta_{in}} \sum_{r \in [R]} \tilde{A}_r(\hat{p}_{rt}^i - \hat{p}_{rt}^{i-1}) \quad (18)$$

According to (16), (17) and (18), we have $P_i - P_{i-1} \geq u_i + \frac{1}{\xi}(D_i - D_{i-1} - u_i)$. Since $\xi \geq 1$ and $u_i \geq 0$, we can further get $P_i - P_{i-1} \geq \frac{1}{\xi}(D_i - D_{i-1})$. $\square$

Theorem 4: TOCA is ξ -competitive, where $\xi = \max_{r \in [R]} \xi_r$ and $\xi_r = \ln(U_r/L_r) + 1, \forall r \in [R]$.

Proof: Since we set $\xi = \max_{r \in [R]} \xi_r$, according to Lemma 3, we have $\hat{p}_{rt}^{i-1}(\beta_{rt}^i - \beta_{rt}^{i-1}) \geq \frac{1}{\xi} \tilde{A}_r(\hat{p}_{rt}^i - \hat{p}_{rt}^{i-1})$. Further, according to Lemma 4, we have $P_i - P_{i-1} \geq \frac{1}{\xi}(D_i - D_{i-1})$.

TABLE II
VM INFORMATION

Type	VM configuration	Computing power
VM1	1vCPUs, 2G RAM, 0.5T HDD	256MIPS
VM2	2vCPUs, 4G RAM, 1T HDD	512MIPS
VM3	4vCPUs, 8G RAM, 2T HDD	1024MIPS
VM4	6vCPUs, 12G RAM, 3T HDD	1536MIPS
VM5	8vCPUs, 16G RAM, 4T HDD	2048MIPS

Next, summing up $P_i - P_{i-1} \geq (D_i - D_{i-1})/\xi$ for all the SMDs' tasks, we can easily get $P_I = \sum_i (P_i - P_{i-1}) \geq \sum_i (D_i - D_{i-1})/\xi = D_I/\xi$. Based on the duality theory [23], we get $D_I \geq V_2$. As a result, we further have $P_I \geq V_2/\xi = V_1/\xi$. Therefore, our proposed TOCA is ξ -competitive. $\square$

V. PERFORMANCE EVALUATION

A. Simulation Settings

We conduct the simulation studies according to the widely-adopted EUA dataset [24], which includes the positions of each SMD and each BS in Metropolitan Melbourne. In the simulations, we consider an MEC system that offers task offloading service within a region of 1000×1000 square meter, in which 20 BSs and 700 SMDs are deployed, and their positions are derived from the EUA dataset. The MEC system lasts over 200 time slots and the duration of each time slot is 10 minutes. The coverage radius of each BS is randomly chosen within [100, 150] meter. According to Google data centers [25], for each BS, the amounts of vCPU, RAM and HDD are randomly drawn from [800, 1000], [1600, 2000] GB and [400, 1000] TB, respectively. Using these resources, each BS provides five types of VMs according to Amazon's EC2, and their specific configuration and computing power are shown in Table II. In addition, the number of subchannels on each BS is randomly picked from [500, 600] and each subchannel owns the bandwidth of 10 Mbps. Each SMD is allowed to apply for up to 5 subchannels. The operational cost of each type of computation and subchannel resources is randomly produced within [0.03, 0.1].

SMDs in the MEC system randomly arrive across the total system lifespan. For each SMD's task, its deadline is uniformly and randomly selected between the arrival time and the system end time, its workload is randomly drawn from (0, 50) Mega Cycles, and its input and output data size are randomly drawn between 50 KB and 1 MB [26], [27]. The transmission power and background noise are randomly generated from [200, 300] mW and $[-100, -50]$ dBm [28], [29]. Based on the physical interference model for wireless communications, we set the channel gain $\sigma_{ij} = D_{ij}^{-\psi}$ where D_{ij} is the distance from SMD i to BS j , and the path loss factor ψ is set as 4 [28], [29]. The bid price of each task offloading scheme is equal to the total consumed resources multiplying unit resource price randomly selected from $[L_r, U_r]$. By default, U_r and L_r are set to 6 and 2, respectively. We repeat each experiment for 100 times to obtain the average results.

Our proposed TOCA is evaluated against two state-of-the-art online auction mechanisms, i.e., ONTA-SWM [30] and OA-CTO [31], and two baseline mechanisms, i.e., RNOM and

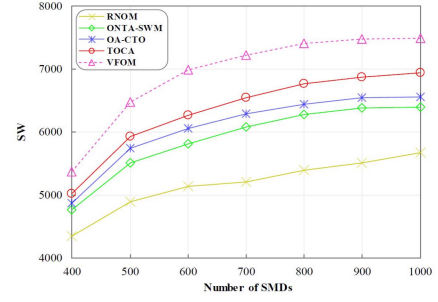


Fig. 2. SW with changed SMDs.

VFOM. Details about ONTA-SWM and OA-CTO can refer to reference [30] and [31], respectively. Upon the arrival of an SMD's task, RNOM randomly chooses a nearby BS to offload its task and randomly schedules the task on this BS. VFOM calculates the optimal offline solution using ILOG CPLEX solver and computes the payment of each winning scheme Π_{in} as follows:

$$p_{in} = \sum_{g \in [I] \setminus \{i\}} \sum_{n \in [N_g]} b'_{gn} x_{gn} - \left(\sum_{g \in [I]} \sum_{n \in [N_g]} b'_{gn} x_{gn} - b'_{in} \right) \quad (19)$$

where $\sum_{g \in [I] \setminus \{i\}} \sum_{n \in [N_g]} b'_{gn} x_{gn}$ is the optimal social welfare that would have been obtained when SMD i had not participated, and $\sum_{g \in [I]} \sum_{n \in [N_g]} b'_{gn} x_{gn} - b'_{in}$ is the sum of all the winning SMD's net valuations except SMD i 's. In addition, since VFOM can obtain the theoretical maximum social welfare, we choose it as the baseline mechanism in order to find the gap between the TOCA solution and the optimal solution.

For a comprehensive performance evaluation, we employ three evaluation metrics, which can reflect different performance attributes of the mechanisms. The evaluation metrics are as follows. 1) Social Welfare (SW): the sum of utilities of the network operator and all the SMDs; 2) SMD Satisfaction (SS): the ratio of the number of winning SMDs to the overall number of SMDs; 3) Resource Utilization (RU): the ratio of the sum of the consumed resources to the overall resources in the MEC system.

B. System Efficiency of TOCA

1) *System Efficiency versus Changed SMDs*: Figs. 2, 3, and 4 respectively show the SW, SS and RU generated by all the involved mechanisms as the number of SMDs increases from 400 to 1000. From Fig. 2, it can be seen that the SW achieved by TOCA is significantly more than that of RNOM, ONTA-SWM and OA-CTO and less than that of VFOM. The main reason is that TOCA considers resource reservation through pricing strategy and enables dynamic resource allocation according to SMDs' task demands and network operator's supplies. Moreover, TOCA supports task offloading in multiple non-consecutive time slots. Noting that the SW of all the involved mechanisms grows as the number of SMDs increases. The reason is that as more SMDs arrive, the network operator

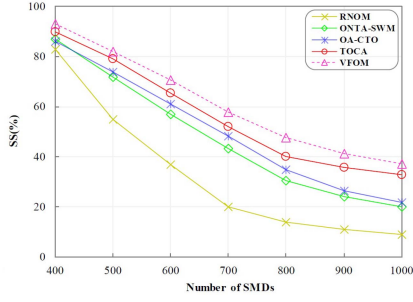


Fig. 3. SS with changed SMDs.

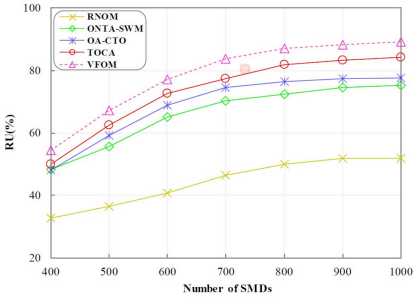


Fig. 4. RU with changed SMDs.

will accept more SMDs until resources are exhausted, which results in the increase in the SW.

From Fig. 3, we can see that the SS of TOCA is obviously higher than that of RNOM, ONTA-SWM and OA-CTO, and quite close to that of VFOM. The reason is that TOCA enables resource reservation, dynamic resource allocation and task offloading in multiple non-consecutive time slots, which leads to more SMDs being admitted. In addition, a higher fraction of SMDs win with a small number of SMDs. This is because the resources are relatively sufficient when the number of SMDs is small.

From Fig. 4, it can be observed that the RU of TOCA is always higher than that of RNOM, ONTA-SWM and OA-CTO, and quite close to that of VFOM. The reason is that TOCA considers resource reservation, dynamic resource allocation, and task offloading in multiple non-consecutive time slots, which enables the network operator to admit more SMDs, and further leads to more resource consumption.

2) *System Efficiency versus Changed BSs*: Figs. 5, 6, and 7 respectively show the SW, SS and RU generated by all the involved mechanisms as the number of BSs increases from 5 to 35. From Fig. 5, we can see that TOCA shows an apparent superiority to RNOM, ONTA-SWM and OA-CTO, since the SW of TOCA is always higher than that of the three involved mechanisms. This is because the resource reservation, dynamic resource allocation and task offloading in multiple non-consecutive time slots in TOCA are more flexible and liable to admit more SMDs in the process of task offloading. In addition, the SW grows with the increasing number of BSs and reaches saturation when there are enough BSs. The reason is that when there are too many BSs, each SMD's task is always offloaded to the most efficient BS,

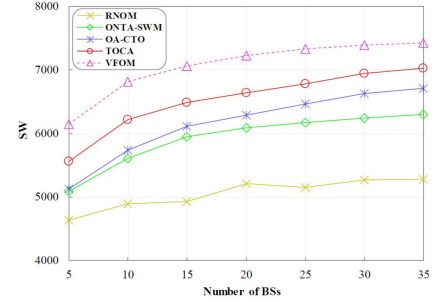


Fig. 5. SW with changed BSs.

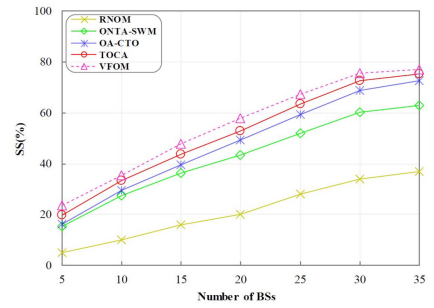


Fig. 6. SS with changed BSs.

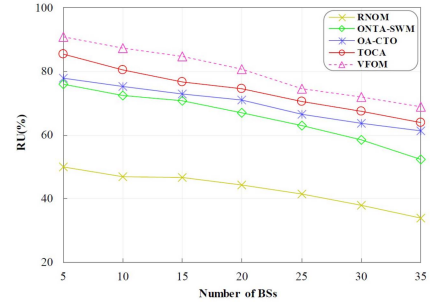


Fig. 7. RU with changed BSs.

while other BSs have no impact on the achievable SW. From Fig. 6, we can find that when the number of BSs increases, the SS achieved by all the involved mechanisms increases. The reason is that with the increasing number of BSs, an increasing amount of resources are available, thus more and more SMDs win, resulting in the increment of the SS. From Fig. 7, we can observe that compared to RNOM, TOCA achieves higher RU in the process of task offloading. ONTA-SWM and OA-CTO are clearly superior to RNOM, but still worse than TOCA. In addition, RU decreases as the number of BSs grows. This is because the consumed resources are significantly less than the increased resources.

3) *System Efficiency versus Changed System Spans*: Figs. 8, 9, and 10 respectively show the SW, SS and RU generated by all the involved mechanisms when system span changes from 50 time slots to 350 time slots. From Fig. 8, we can see that the SW increases with the increasing length of the system span. The reason is that TOCA enables resource reservation, dynamic resource

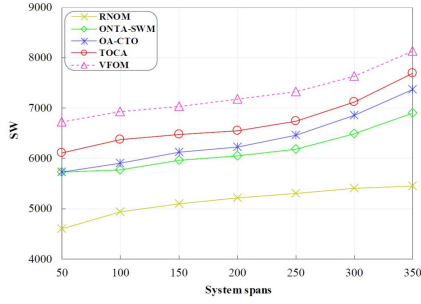


Fig. 8. SW with changed system spans.

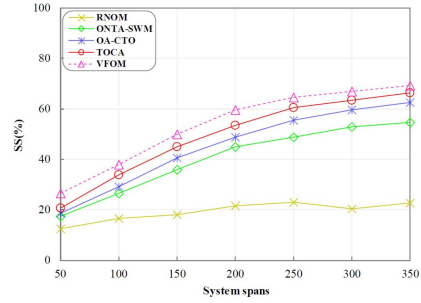


Fig. 9. SS with changed system spans.

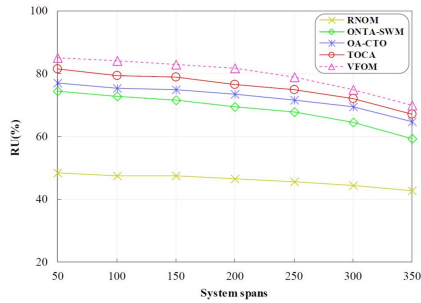


Fig. 10. RU with changed system spans.

allocation, and task offloading in multiple non-consecutive time slots, resulting in admitting more SMDs with the increasing length of the system span. Fig. 9 reflects the SS of all the involved mechanisms gradually rises when the length of system span increases. The reason is that the probability of winning becomes higher when the length of system span increases, since there exist more time slots available for the task scheduling. From Fig. 10, we can see that the RU of all the involved mechanisms exhibits a declining trend when the length of system span increases. The reason is that an increase in system span is equivalent to an increase in the amounts of resources, meanwhile, the consumed resources are significantly less than the increased resources.

C. Property Verification

1) *Competitive Ratio*: Figs. 11, 12, and 13 respectively show the competitive ratios under different number of SMDs, different number of BSs and different length of system spans. From

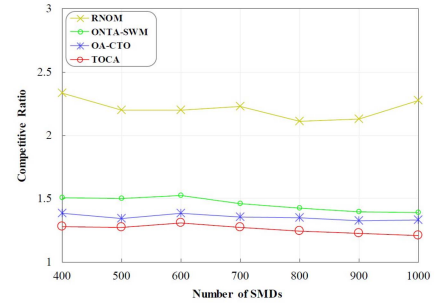


Fig. 11. Competitive ratio with changed SMDs.

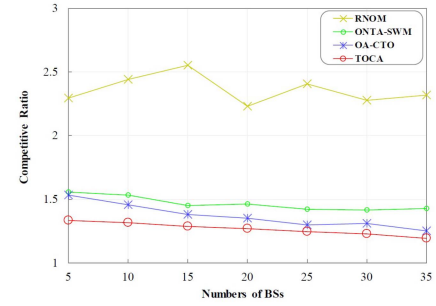


Fig. 12. Competitive ratio with changed BSs.

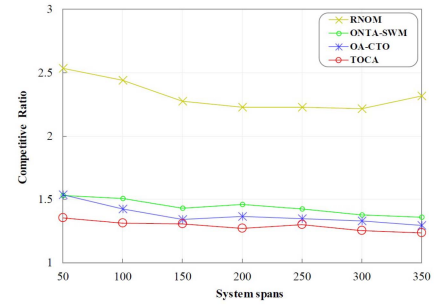


Fig. 13. Competitive ratio with changed system spans.

Fig. 11, we can see that the actual competitive ratio of TOCA is less than 1.5, which is rather lower than the theoretical upper bound. It fluctuates when the number of SMDs grows, indicating that the number of SMDs has no main impact on the competitive ratio, which is in line with the theoretical analysis. From Fig. 12, we can see that the competitive ratios of TOCA slightly decline as the number of BSs increases. The reason is that more and more BSs can offer more resources to satisfy the SMDs' requests and increase the number of candidate BSs around each SMD. From Fig. 13, we can see that the competitive ratio of TOCA is also less than 1.5, which is in line with the theoretical analysis. Moreover, the competitive ratio almost remains unchanged with different length of system span, which indicates that the our proposed TOCA is stable.

2) *Truthfulness*: In order to verify the truthfulness of TOCA, we randomly choose a winning SMD's bid $B_{i'}$ and a losing SMD's bid $B_{i''}$, to observe how their utilities vary with the changed bid prices. The results are shown in Fig. 14.

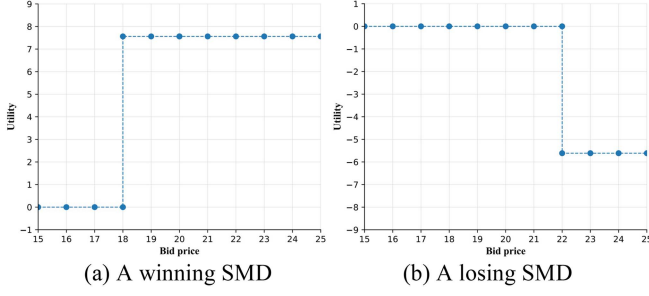


Fig. 14. Truthfulness.

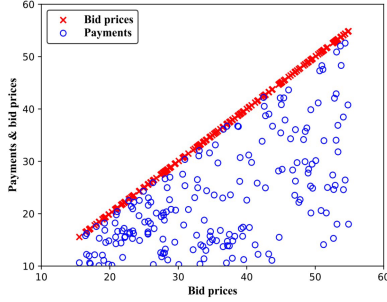


Fig. 15. Individual rationality.

TABLE III
RUNNING TIME (MS)

$I/J/T$	600/15/150	700/20/200	800/25/250	900/30/300	1000/35/350
RNOM	2.41×10^1	3.58×10^1	5.24×10^1	8.22×10^1	6.25×10^2
ONTA-SWM	5.14×10^1	7.48×10^1	8.61×10^1	4.77×10^2	1.64×10^3
OA-CTO	6.76×10^1	8.45×10^1	9.58×10^1	6.24×10^2	3.73×10^3
TOCA	9.30×10^1	1.28×10^2	2.26×10^2	8.72×10^2	6.24×10^3
VFOM	1.55×10^4	8.87×10^4	3.06×10^5	1.21×10^6	2.47×10^7

From Fig. 14(a), it can be seen that when the bid price $b_{i'}$ is equal to its real valuation, i.e., $b_{i'} = v_{i'} = 25.6$, SMD i' wins and obtains the utility of 7.6. It is clear that no matter how the winning SMD announces its bid price, the utility cannot be improved. From Fig. 14(b), we can see that when the bid price $b_{i''}$ is equal to its real valuation, i.e., $b_{i''} = v_{i''} = 16.3$, it does not win and obtains the utility of 0. It is also clear that the utility of $B_{i''}$ will be no more than 0, even if the losing SMD reports untruthful valuation.

3) *Individual Rationality*: To verify the individual rationality of TOCA, the payments and the bid prices of the winning bids are plotted in Fig. 15. It is clear that the payment of each winning bid is always less than the corresponding bid price. Therefore, TOCA achieves individual rationality.

4) *Computational Efficiency*: To examine computational efficiency of TOCA, we compare the running time of TOCA with that of RNOM, ONTA-SWM, OA-CTO and VFOM, as shown in Table III.

From Table III, we can see that TOCA executes much faster than VFOM, which exponentially increases, and slightly slower

than RNOM, ONTA-SWM, OA-CTO. Note that the growth rate of the running time vs the scale in TOCA is linear, which is in line with Theorem 3. Moreover, at the cost of computation overhead, TOCA can achieve higher SW with resource reservation, dynamic resource allocation and task offloading in multiple non-consecutive time slots.

VI. RELATED WORKS

Inspired by the Amazon cloud auctions, there exist many works focusing on designing auction-based mechanisms for cloud computing markets, where heterogeneous VM instances are requested by users [32], [33], [34], [35], [36], [37]. Nevertheless, these auction-based mechanisms cannot directly be applicable to an MEC system, in which much more complicated constraints need to be considered, such as locality constraint, delay constraint, etc. Therefore, a number of auction-based mechanisms [26], [27], [28], [30], [31], [38], [39], [40], [41], [42], [43], [44], [45], [46] have been deliberately designed for MEC recently.

In [26], a repeated auction mechanism is presented for load-aware dynamic resource allocation in an MEC system, which achieves computational efficiency, individual rationality, and acquires a symmetric Nash Equilibrium solution. In [38], two resource allocation and pricing mechanisms are proposed for MEC systems, in which users have different requests and compete for high quality services. Nevertheless, the mechanisms in [26] and [38] cannot achieve truthfulness of SMDs, which may not only cause harm to other SMDs, but also indirectly lead to profit losses for the network operator.

In [28], a truthful combinatorial auction mechanism is presented to allocate computation and subchannel resources jointly while satisfying the constraints of energy and delay of the corresponding tasks. In [39], two double auction mechanisms with dynamic pricing are proposed for MEC systems to decide the matching pairs between SMDs and BSs and their corresponding payments under the locality constraints. In [40], a novel service-oriented double auction mechanism is proposed to solve the problems of incentive design and service provisioning jointly in MEC systems. In [41], a truthful combinatorial double auction mechanism is presented, which introduces the padding method and the efficient pricing scheme to achieve truthfulness in MEC systems. In [42], a truthful auction-based resource allocation mechanism with flexible task offloading is proposed for an MEC system, aiming to maximize the SW while achieving truthfulness and satisfy the locality constraint. In [43], a reverse auction-based task offloading and resource allocation mechanism is presented to encourage SBSs to take part in the process of task offloading. In [44], a latency-sensitive double auction-based resource allocation mechanism is proposed for the market-oriented MEC systems. Nevertheless, the mechanisms proposed in [28], [39], [40], [41], [42], [43], [44] focus on system performance optimization in offline scenario, ignoring the online scenario in which tasks arrive on the fly, which is a typical scenario in MEC systems and brings more challenges in the system performance optimization.

TABLE IV
COMPARISON OF TOCA WITH EXISTING AUCTION MECHANISMS

Ref.	OL	JCC	MSMB	T	MTS	NCTS	DC	LC	ABSD	DRA	SWM
[26]	×	×	√	×	×	×	√	×	×	√	×
[27]	√	√	√	×	√	×	√	×	×	×	×
[28]	×	√	√	√	×	×	√	×	×	√	√
[30]	√	×	√	√	√	×	×	×	×	√	√
[31]	√	√	×	√	√	×	√	√	×	√	√
[38]	×	×	√	×	×	×	×	×	×	√	√
[39]	×	×	√	√	×	×	×	×	×	√	×
[40]	×	√	√	√	×	×	×	×	×	√	×
[41]	×	×	√	√	×	×	×	√	√	√	√
[42]	×	√	√	√	×	×	√	√	√	√	√
[43]	×	√	√	√	×	×	√	×	×	√	×
[44]	×	√	√	√	×	×	√	×	×	×	×
[45]	√	√	√	√	√	×	×	×	×	√	×
[46]	√	√	√	√	×	×	√	×	×	√	×
TOCA	√	√	√	√	√	√	√	√	√	√	√

Online (OL), joint computation and communication resource allocation (JCC), multipleSMDsand multiple BSs (MSMB), truthfulness (T), task offloading in multiple time slots (MTS), task offloading in non-consecutive time slots (NCTS), deadline constraint (DC), locality constraint (LC), adjacent BS determination (ABSD), dynamicresourceallocation (DRA), social welfare maximization (SWM).

Several online auction mechanisms [27], [30], [31], [45], [46] are designed for MEC systems. In [27], an online reverse auction-based task assignment and scheduling mechanism is presented to select a set of tasks for auctions and to determine the task executor for each selected task. Nevertheless, the mechanism in [27] cannot guarantee the truthfulness and assumes that the requesters' resource demands are homogeneous and each task executor can process only one task at a time. In [30], an online incentive mechanism is designed for the MEC system, which performs decision-making only according to the current resource request information. However, the mechanism in [30] doesn't consider the constraints of deadline and locality, and ignores the subchannel allocation. In [31], a truthful online auction-based mechanism is proposed to maximize the social welfare of the MEC system. However, it cannot be directly applied to our considered scenario, in which there are multiple BSs and multiple SMDs and a task can be offloaded in multiple non-consecutive time slots. Moreover, it lacks of the adjacent BS determination. In [45], an online multi-round auction mechanism is designed to trade the resources between BSs and SMDs, aiming to pursue BSs' profit maximization. However, it doesn't consider the constraints of locality and deadline, and ignores the interests of SMDs. In [46], an online rewards-optimal auction is presented to maximize the sum of long-term rewards of the MEC system with explicit consideration of the highly dynamic energy harvesting process. However, the mechanism in [46] ignores the locality constraint and the interests of IoT devices. Moreover, it requires that a computation task should be processed within one time slot, which may be unrealistic in an MEC system. To better understand of the differences between the proposed TOCA and the existing auction-based mechanisms, the comparative study is listed in Table IV.

From Table IV, we can see that most of the existing auction-based mechanisms for MEC task offloading do not simultaneously consider the constraints of locality and deadline in

the online scenario. Moreover, they cannot support dynamic computation and subchannel resource allocation for task execution in multiple non-consecutive time slots. Consequently, it is necessary to devise effective mechanisms to bridge the gap between the existing approaches and the task offloading problem in MEC systems.

VII. CONCLUSION

In this paper, we present a truthful online combinatorial auction-based mechanism (TOCA) for task offloading in multiple non-consecutive time slots, which enables dynamic computation and communication resource allocation with explicit consideration of the constraints of deadline, locality and resource. Specifically, we first devise the *candidate offloading scheme determination*, aiming to determine the candidate offloading schemes of an SMD upon the arrival of its task. Next, we devise the *winning offloading scheme selection and pricing* based on the online primal-dual optimization framework, to decide the winning scheme among the SMD's candidate offloading schemes and calculate its corresponding payment. Theoretical analysis proves that TOCA can achieve truthfulness, individual rationality, computational efficiency and a good competitive ratio. Trace-driven simulation results verify the effectiveness and efficacy of the proposed TOCA.

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